ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

53002052 - Mercados Ambientales Y De Energías Renovables

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	53002052 - Mercados Ambientales y de Energías Renovables
No of credits	3 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Carlos Enrique Vazquez Martinez (Subject coordinator)	518	vazquez.martinez@upm.es	Tu - 11:00 - 13:00 W - 11:00 - 13:00 Th - 11:00 - 13:00 Please, request an appointment by email

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Economía De La Energía

3.2. Other recommended learning outcomes

The subject - other recommended learning outcomes, are not defined.

4. Skills and learning outcomes *

4.1. Skills to be learned

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CE11 - Analizar el comportamiento energético y control de los sistemas de energías renovables determinando y aplicando criterios innovadores de optimización energética, económica y ambiental, aplicando metodologías de diseño, simulación y análisis de los componentes y sistemas de energías renovables: solares, eólicos, hidráulicos, de biomasa, de energías marinas, geotérmicas y otras energías renovables; para contribuir a su desarrollo tecnológico y a su competitividad con otras tecnologías energéticas.

CE13 - Entender la evolución y el funcionamiento de los mercados de petróleo, gas y electricidad. Conocer los principales tipos de diseño de los mercados de electricidad y gas que existen en la experiencia internacional y los criterios bajo los que se han diseñado, y ser capaz de analizar cuál es la regulación más adecuada para cada situación.

CE18 - Entender la optimización de costes en una empresa: coste marginal, coste medio, coste hundido, coste de oportunidad, aplicados al sector de la energía. Analizar costes en el sector de la energía.

CE4 - Comprender y aplicar los principios de funcionamiento, formación de precios y equilibrio en los mercados energéticos, tanto en condiciones de competencia perfecta como en condiciones de competencia imperfecta

CE6 - Disponer de habilidades, criterios y conocimientos para analizar de forma objetiva el impacto ambiental de cualquier fuente de energía.

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG2 - Poseer capacidad para diseñar, desarrollar, implementar, gestionar y mejorar productos, sistemas y procesos en los distintos ámbitos energéticos, usando técnicas analíticas, computacionales o experimentales avanzadas.

CT1 - Aplica. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

CT10 - Conoce. Conocimiento de los temas contemporáneos.

CT11 - Usa herramientas. Habilidad para usar las técnicas, destrezas y herramientas ingenieriles modernas necesarias para la práctica de la ingeniería.

CT3 - Diseña. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

CT4 - Trabaja en equipo. Habilidad para trabajar en equipos multidisciplinares.

CT5 - Resuelve. Habilidad para identificar, formular y resolver problemas de ingeniería.

CT7 - Comunica. Habilidad para comunicar eficazmente.

CT9 - Se actualiza. Reconocimiento de la necesidad y la habilidad para comprometerse al aprendizaje continuo.

4.2. Learning outcomes

RA168 - Analizar y entender los diferentes tipos de diseño de mercado de los mecanismos de apoyo a la inversión de generación eléctrica, convencional y renovable

RA136 - Conocer la situación actual, evolución hasta el momento y perspectivas futuras de los diferentes mercados de energía

RA205 - Analizar y entender los diferentes diseños de mecanismos de apoyo a la generación de electricidad a partir de fuentes renovables

RA206 - Analizar y entender los diferentes diseños de métodos de reducción de emisiones contaminantes

RA207 - Analizar y entender los diferentes diseños de mecanismos regulatorios de fomento de la eficiencia y el ahorro energéticos, incluyendo los mercados de flexibilidad y el almacenamiento de energía

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The course focuses on the study of the environmental policies that have played a key role in the evolution of energy systems around the world over the last decade. The analysis will be structured around three broad topics: a) the different regulatory mechanisms available to address carbon emission reduction; b) the support systems oriented to renewable electricity generation; and c) the mechanisms devised to promote energy efficiency and, more generally, behind-the-meter technologies.

5.2. Syllabus

1. Introduction
2. Carbon emissions markets
3. Generation investment
    3.1. Energy-only markets
    3.2. Conventional technologies
    3.3. Renewable technologies
    3.4. General designs
4. Behind-the-meter
    4.1. Market design for energy efficiency and distributed generation
    4.2. Flexibility markets
    4.3. Energy poverty

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Introduction** Duration: 02:00 Lecture			
2	**Follow-up test** Duration: 00:05 Additional activities **Carbon markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
3	**Follow-up test** Duration: 00:05 Additional activities **Carbon markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
4	**Follow-up test** Duration: 00:05 Additional activities **Carbon markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
5	**mid-term exam** Duration: 00:50 Additional activities **Investment in generation: statement of the problem** Duration: 01:10 Lecture			**Mid-term exam** Written test Progressive assessment Presential Duration: 00:50
6	**Follow-up test** Duration: 00:05 Additional activities **Investment in generation: energy-only markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
7	**Follow-up test** Duration: 00:05 Additional activities **Investment in generation: conventional generation** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05

8	**Follow-up test** Duration: 00:05 Additional activities **Investment in generation: renewable generation** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
9	**Follow-up test** Duration: 00:05 Additional activities **Investment in generation: modern auctions** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
10	**mid-term exam** Duration: 00:50 Additional activities **Market design for promoting behind-the-meter actions** Duration: 01:10 Lecture			**Mid-term exam** Written test Progressive assessment Presential Duration: 00:50
11	**Follow-up test** Duration: 00:05 Additional activities **Market design for promoting behind-the-meter actions** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
12	**Follow-up test** Duration: 00:05 Additional activities **Market design for promoting behind-the-meter actions** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
13	**Follow-up test** Duration: 00:05 Additional activities **Flexibility markets and energy poverty** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
14	**Flexibility markets and energy poverty** Duration: 01:10 Lecture **mid-term exam** Duration: 00:50 Additional activities			**Mid-term exam** Written test Progressive assessment Presential Duration: 00:50

15				
16				
17				**Final exam** Written test Progressive assessment Presential Duration: 02:00 **Final exam** Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
2	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
3	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

4	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
5	Mid-term exam	Written test	Face-to-face	00:50	25%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
6	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

7	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
8	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
9	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

10	Mid-term exam	Written test	Face-to-face	00:50	25%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
11	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
12	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

13	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
14	Mid-term exam	Written test	Face-to-face	00:50	25%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18
17	Final exam	Written test	Face-to-face	02:00	75%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE6 CE11 CE13 CE18

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen extraordinario	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4

						CE6 CE11 CE13 CE18

7.2. Assessment criteria

For students taking the course through progressive assessment,

- 25% of the final grade corresponds to the follow-up tests. There is no minimum grade for each individual test, but an average grade of 4 is required. Also, it is required to take at least 70% of the follow-up tests. Otherwise, the student will have to take the whole final exam.
- 75% of the final grade corresponds to the three mid-term exams, each of them weighting 25%. A minimum grade of 5 is required in each of the midterm exams. In case any of the midterm exams fails under that minumum grade, the topics corresponding to that exam will have to be evaluated again at the final examination. The topics covered by midterm exams with grades over 5 will not have to be evaluated again at the final exam. The relevant grade for calculating the global average will be the last exam taken, for each part.

For students taking the course under the "final exam only" option,

- 100% of their final grade will be awarded for the final exam

When an extraordinary exam is required, the entire syllabus must be evaluated, and 100% of the grade will be awarded for the extraordinary exam

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
I.J. Pérez-Arriaga (Editor), "Regulation of the Power Sector", Springer, 2013	Bibliography	Power markets, focus on market design
S. Hunt, "Making Competition Work in Electricity", 2002	Bibliography	Power markets, focus on competition
www.irena.org	Web resource	Renewable energy. Focus on costs rather than markets

9. Other information

9.1. Other information about the subject

Students are encouraged to take this course simultaneously with the Electricity Markets course, as both are related and topics are interlinked.
This course is related to Sustainable Development Goal number 7: affordable and clean energy